Questions

Q1.

The International Space Station (ISS) has several solar panels called wings.

(a) The wings convert energy from the Sun into a form useful in the ISS.

(1)

- ☐ **A** transverse and electromagnetic
- ☐ **B** electromagnetic but not transverse
- ☐ **C** transverse but not electromagnetic
- ☐ **D** neither transverse nor electromagnetic

(b) In one second, the useful energy available from one wing is 34.3 kJ.
The energy incident on the wing from the Sun is five times this amount.

What is the percentage efficiency of the wing?

(3)

efficiency = ... %

(c) A wing is in direct sunlight.
The ISS is not receiving energy from the wing.
The temperature of the wing remains constant.

Explain why the temperature of the wing remains constant in these conditions.

(2)

.....

.....

.....

.....

Q2.

(a) Here are some forms of energy:

chemical	elastic potential	electrical
heat (thermal)	kinetic	light
nuclear	sound	

(i) Use words from the box to complete the table.
Each word may be used once, more than once, or not at all.

The first one has been done for you.

(3)

device	energy transferred from...	energy is mostly transferred into...
electric motor	electrical	kinetic
bow and arrow	elastic potential	
electric kettle	electrical	
microphone		electrical

(ii) In the electric motor only some of the electrical energy is transferred into kinetic energy.

State what happens to the remaining electrical energy.

(1)

.....

(b) Many appliances are sold with an energy efficiency rating.
A-rated appliances are the most energy efficient.

Here is some information about two types of electric lamp.

	halogen lamp	compact fluorescent lamp (CFL)
energy efficiency rating	B	A
energy transfer diagrams (not drawn to scale)	energy transfer in one second electrical energy = 20 J light energy = 2 J	energy transfer in one second electrical energy = 20 J light energy = 8 J

(i) Calculate how much energy is wasted in one second by the compact fluorescent lamp (CFL).

(1)

.....

(ii) Use the energy transfer diagrams to explain why the CFL lamp has a better efficiency rating than the halogen lamp.

(2)

.....

.....

.....

.....

(c) The photograph shows an electric heater used to warm garages.

When the heater is switched on, it quickly warms up and then stays at a constant temperature. Explain why the heater stays at a constant temperature.

(2)

.....

.....

.....

.....

(Total for Question is 9 marks)

Q3.

A student wants to use sunlight to heat water in a container.

(i) The student can choose from four containers.

The containers:

- are all the same size
- have different colours
- have different surfaces.

Which container is the best absorber of thermal energy?

Put a cross (☒) in the box next to your answer.

(1)

	container colour	container surface
☐ A	light	shiny
☐ B	light	rough
☐ C	dark	shiny
☐ D	dark	rough

(ii) The student places the container in bright sunlight. After half an hour, the temperature of the water is 50°C. The water then stays at 50°C, even though the container is still in bright sunlight.

Which of these explains why the temperature stays at 50°C?
Put a cross (☒) in the box next to your answer.

(1)

☐ **A** The water in the container is no longer absorbing thermal energy.

☐ **B** The container is losing thermal energy faster than it is absorbing it.

☐ **C** The container is losing thermal energy at the same rate it is absorbing it.

☐ **D** The water in the container has started to boil.

Q4.

The photograph shows an electric heater used to warm garages.

When the heater is switched on, it quickly warms up and then stays at a constant temperature.
Explain why the heater stays at a constant temperature.

(2)

.....

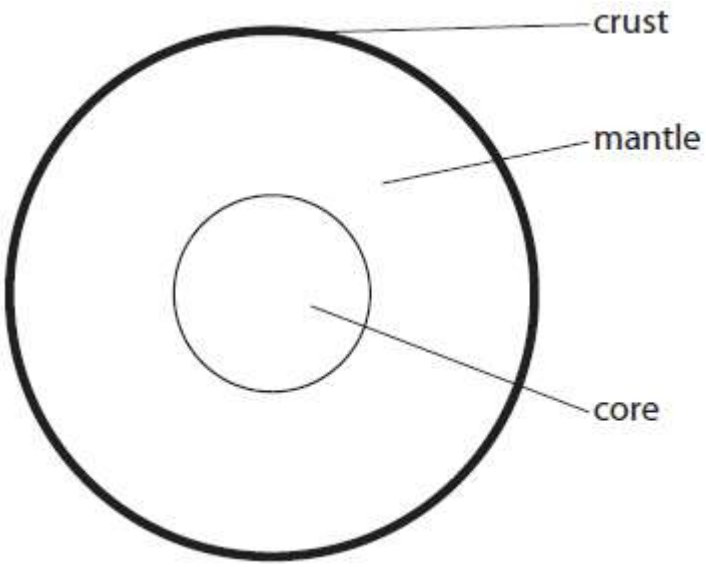
.....

.....

.....

Q5.

The diagram shows a cross-section of the Earth.



The mantle is hotter near the core than near the crust.

(i) Complete the sentence by putting a cross (☒) in the box next to your answer.

One reason for the mantle being hotter near the core is that

(1)

☐ **A** the Earth's crust is a solid

☐ **B** the Earth's core is a liquid

☐ **C** the Earth is radiating heat to space

☐ **D** the Earth is absorbing heat from space

(ii) Explain how this temperature difference causes the tectonic plates in the Earth's crust to move.

(2)

.....

.....

.....

.....

Q6.

Figure 1 is a ray diagram for a converging lens when used as a magnifying glass.

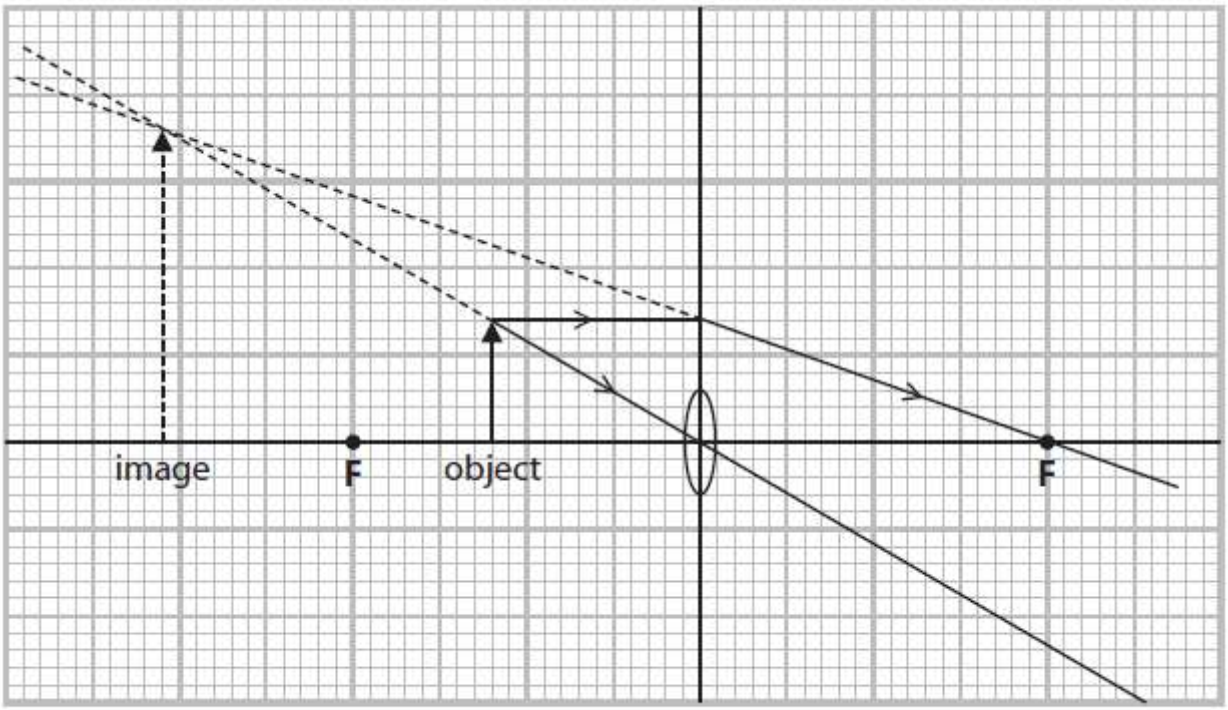


Figure 1

(i) Using information from Figure 1, determine the magnification of the virtual image.
Use the equation

$$\text{magnification} = \frac{\text{height of image}}{\text{height of object}}$$

(2)

magnification = ...

(ii) Describe **one** way the magnification of the image could be increased.

(2)

.....

.....

.....

.....

(Total for question = 4 marks)

Q7.

Microwaves are used to send and receive television signals from high orbit geostationary satellites.

Describe why microwaves rather than radio waves are used.

(2)

.....

.....

.....

.....

(Total for question = 2 marks)

Q8.

A student investigates what happens when a ray of light shines into a semicircular block made of transparent plastic.

The ray meets the straight edge at an angle of 20°, as shown in Figure 13.

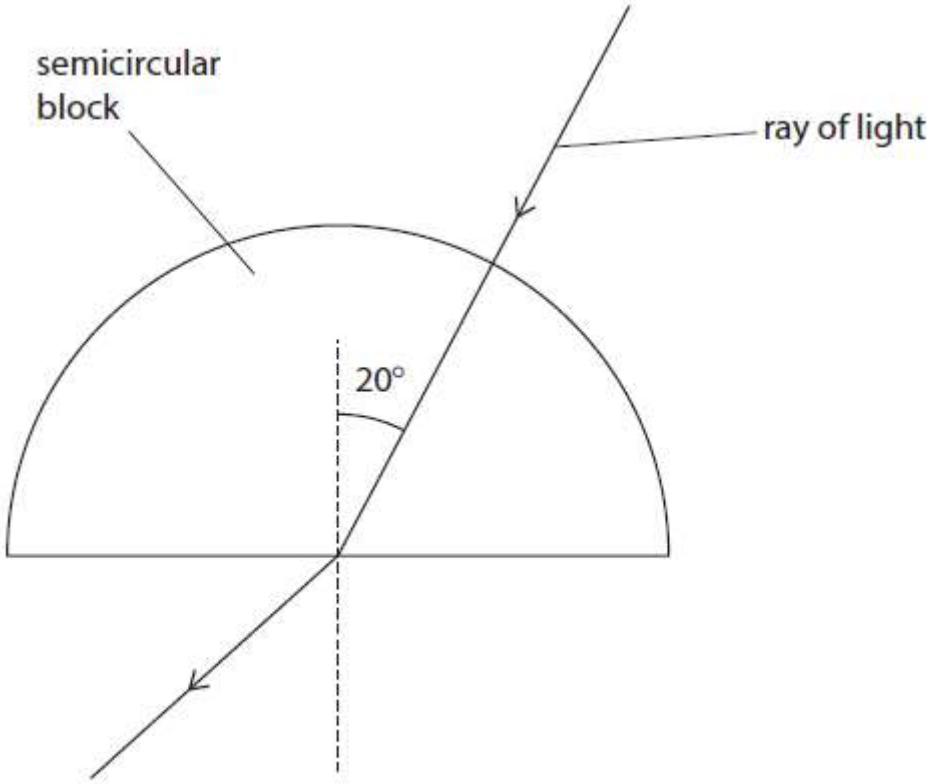


Figure 13

The student gradually increases the angle at the straight edge from 20° to 50° and observes what happens to the ray of light.

The critical angle for the transparent plastic is 38°.

Describe what you would expect to happen to the ray of light as the angle at the straight edge increases from 20° to 50°.

You may add to the diagram in Figure 13 to help your answer.

(3)

.....

.....

.....

.....

.....

.....

.....

(Total for question = 3 marks)

Q9.

A student investigates how the angle of refraction varies with the angle of incidence when light enters a glass block.

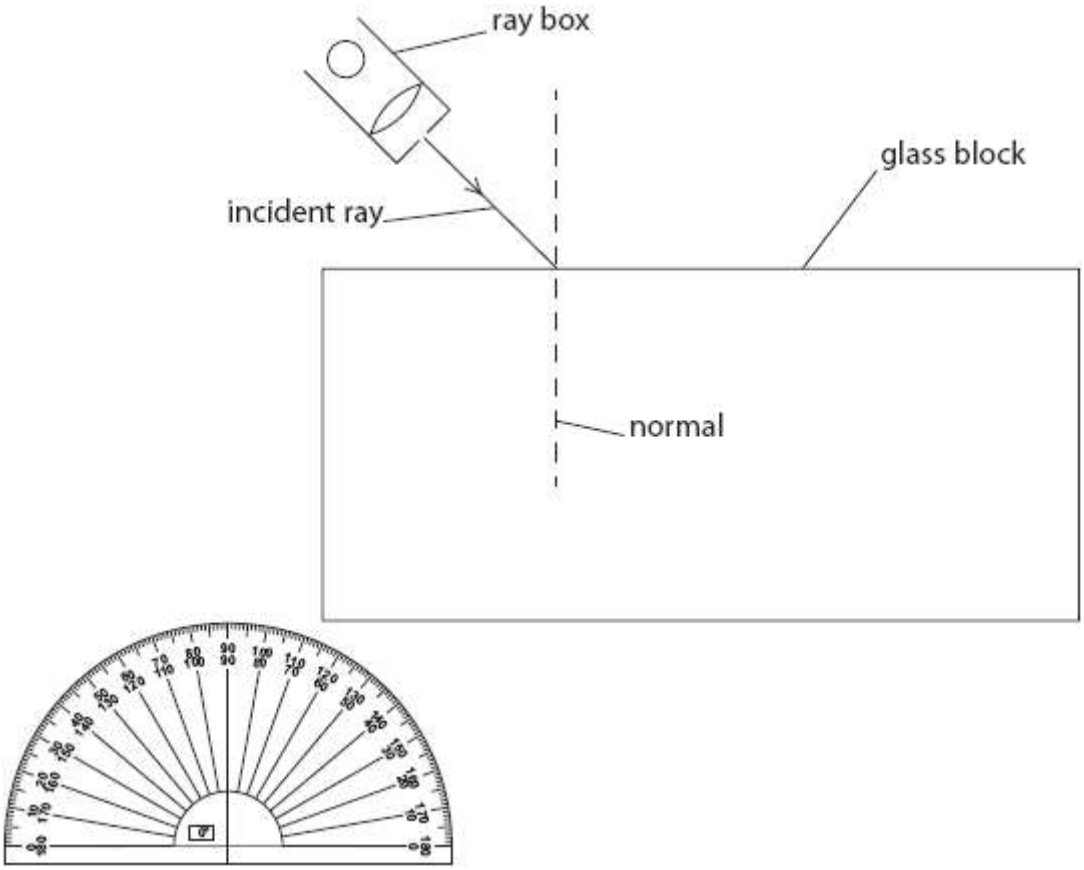


Figure 1

Describe how the student should perform this investigation.

(4)

.....

.....

.....

.....

.....

.....

.....

(Total for question = 4 marks)

Q10.

Figure 3 shows a semicircular glass block.

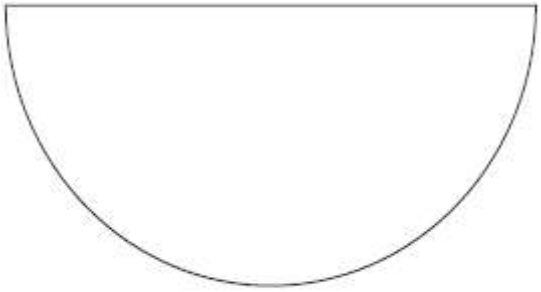


Figure 3

Describe how a student could use the semicircular glass block and other apparatus to determine the critical angle for a glass-air boundary.

You should add to the diagram in Figure 3 to help with your answer.

(4)

.....

.....

.....

.....

.....

.....

.....

.....

(Total for question = 4 marks)

Mark Scheme

Q1.

Question Number	Answer	Acceptable answers	Mark
(a)	A transverse and electromagnetic		(1)

Question Number	Answer	Acceptable answers	Mark
(b)	Evaluation 171.5 (1) Substitution (34.3/171.5) x 100 (1) Evaluation 20 (%) (1)	award full marks for correct answer with no working 34.3 x 5 [34.3 /(34.3 x 5)] x 100 [34.3 /(34.3 x 5)] [34.3 /171.5] Allow 0.2 or 1/5 for 3 marks	(3)

Question Number	Answer	Acceptable answers	Mark
(c)	rate of {energy/heat} (from the Sun){absorbed/taken in} (1) equals rate of {energy/heat} {radiated/emitted/given out}(1)	Allow 'energy in = energy out' for 1 mark 'power in = power out' for 2 marks	(2)

Q2.

	Answer	Acceptable answers	Mark
(ai)	(Bow and arrow:) kinetic (1) (Electric kettle:) heat (thermal) (1) (Microphone:) sound (1)	Heat/thermal	**(3)**
(a)(ii)	Any **one** from (transferred into) thermal/heat/sound(energy) (1) (Energy) is dissipated (1)	Do not accept light energy or it disappears goes into surroundings/air (energy) is wasted/lost	**(1)**
(b)(i)	12 (J) Ignore any unit given by candidate.	20 - 8 (J)	**(1)**
(b)(ii)	An explanation linking any **two** of • (For the) same	Same input (energy) gives out/produces more	**(2)**

	amount of electrical/supplied (energy/power) (1) • (CFL/it) has a greater output (of light energy) (1) • (CFL/it) wastes less (electrical energy) (1)	light/useful (energy) Do not accept more energy is used in the (CFL/it) Ignore brightness. (CFL/it) produces less thermal/heat (energy) Accept explanations using data from the energy transfer diagrams as comparisons eg (CFL/it) is four times as efficient gains both marks	
(c)	An explanation linking • dissipating heat (1) • at same (rate)/as quickly as energy is being supplied (1)	gives out/radiates/conducts/ convects /loses /produces heat/thermal/ energy gives out as much energy/power as it takes in(each second) Gains both marks If no other marks scored: There is a constant current/ steady flow of energy into the heater gains one mark Ignore refs to thermostat	**(2)**

Total for Question = 9 marks

Q3.

Question Number	Answer	Acceptable answers	Mark
(i)	**D** dark : rough		**(1)**

Question Number	Answer	Acceptable answers	Mark
(ii)	**C** the container is losing thermal energy at the same rate it is absorbing it	Heat for thermal Same amount in same time for same rate	**(1)**

Q4.

	Answer	Acceptable answers	Mark
	An explanation linking • dissipating heat (1) • at same (rate)/as quickly as energy is being supplied (1)	gives out/radiates/conducts/ convects /loses /produces heat/thermal/ energy gives out as much energy/power as it takes in(each second) Gains both marks If no other marks scored: There is a constant current/ steady flow of energy into the heater gains one mark Ignore refs to thermostat	**(2)**

Q5.

Question Number	Answer	Acceptable answers	Mark
(i)	**C** The Earth is radiating heat to space		**(1)**

Question Number	Answer	Acceptable answers	Mark
(ii)	An explanation linking any 2 of: • Hot material rises/cold material falls (1) • (causes) material under the plate to move sideways (1) • (because of) uneven heating (1)	convection current in the {mantle/magma/under plates} heat from core warms mantle near core IGNORE temperature difference (in stem)	**(2)**

Q6.

Question Number	Answer	Additional guidance	Mark
(i)	substitution (1) $\frac{18.0 \pm 1}{7.0 \pm 1}$ **OR** $\frac{3.5 \pm 0.2}{1.4 \pm 0.2}$ evaluation (1) 2.6 ± 0.5	award full marks for correct answer without working	**(2)** **AO2**

Question Number	Answer	Additional guidance	Mark
(ii)	a description to including one pair of: use different lens/replace lens (1) {higher power / shorter focal length / fatter / thicker / more curved / greater refractive index / more convex} lens (1) **OR** change the distance / move the lens / move the object (1) further away (from the object/lens) (1)	accept stronger curvature of lens increased (2) increase distance between object and lens (2)	**(2)** **AO3**

Q7.

Question number	Answer	Additional guidance	Mark
	An answer that combines the following points of understanding to provide a logical description: • radio waves will not reach the satellites (from Earth) / be received (on Earth) from the satellites (1) • because they are reflected by the atmosphere (1)	ORA for microwaves reflected by ionosphere / before reaching satellite	**(2)**

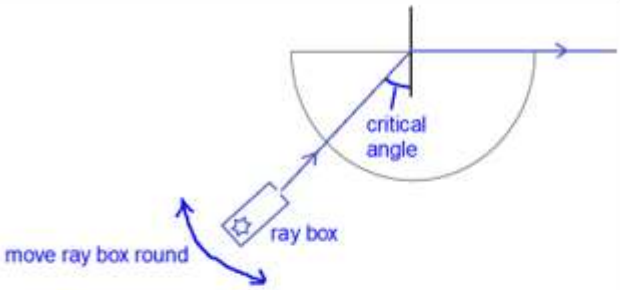
Q8.

Question number	Answer	Additional guidance	Mark
	description to include **three** from	ignore partial reflections marks can be gained from labelled diagram	**(3)** **AO2.2**
	when the angle inside (the block) increases the angle outside the block increases (1)	outside ray (moves/bends) away from the normal OR towards the block refract mores	
	when the angle inside the block = 38° the angle outside the block = 90° (1)	allow critical angle for 38 the ray goes along the straight edge	
	when the angle inside the block > 38° ray reflected back inside (1)	total internal reflection or TIR just after 38°	
	when the angle inside the block > 38° angle i = angle r OWTTE (1)		

Q9.

Question number	Answer	Additional guidance	Marks
	An answer that that combines the following to provide a logical description of the method	Responses may be seen on diagram. allow 'calculate' for 'measure' in context ('measure angles' insufficient – must be specified)	(4)
	• measure angle of incidence (1)	measure angle from incident ray	
	• measure angle of refraction (1)	measure angle from refracted ray accept (reference to drawing path of) refracted ray (inside block) / emergent ray / ray under or leaving block	
	• measure angle(s) to the **normal** (1)	allow to 'dotted line' for normal	
	• repeat for different angle(s) (of incidence) (1)		

Q10.

Question Number	Answer	Additional guidance	Mark
	a description to include any **four** from:		**(4)**
	shine a ray (of light) into the block (1)		
	into block through the curved face along a radius (1)	credit marking points in the diagram if they are clear	
	{change angle / move ray(box)} until {the angle of refraction is 90°/ TIR just occurs} (1)		
	measure angle of incidence {when refracted angle is 90° / when TIR just occurs} (1)	allow 'calculate' for 'measure' plot angle i against angle r	
	repeat measurement of critical angle (1)	if light only enters block at straight edge, maximum 1 mark (for MP1)	